

Machine Learning

CSE 4621

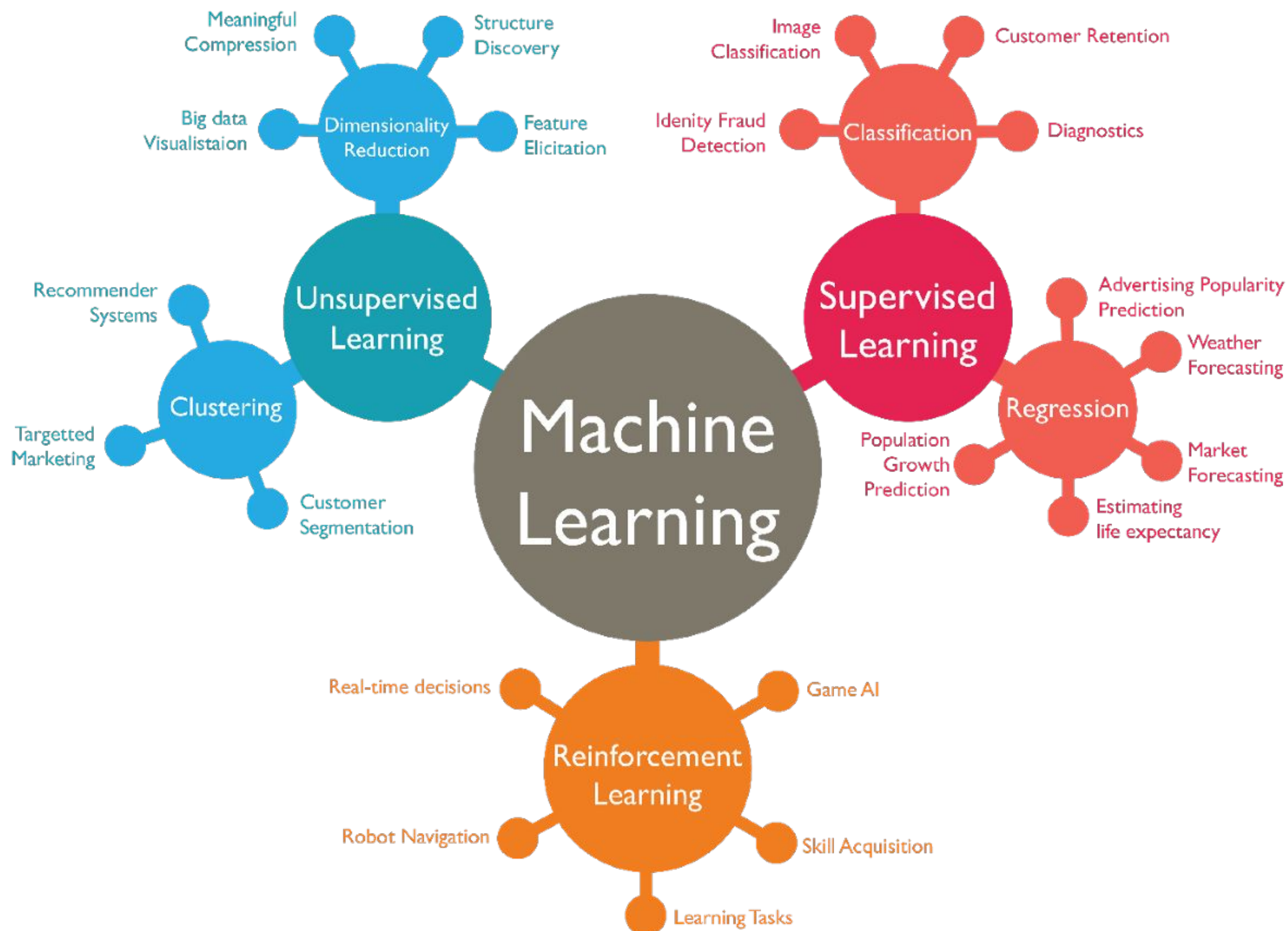
Presented by

Rafsanjany Kushol

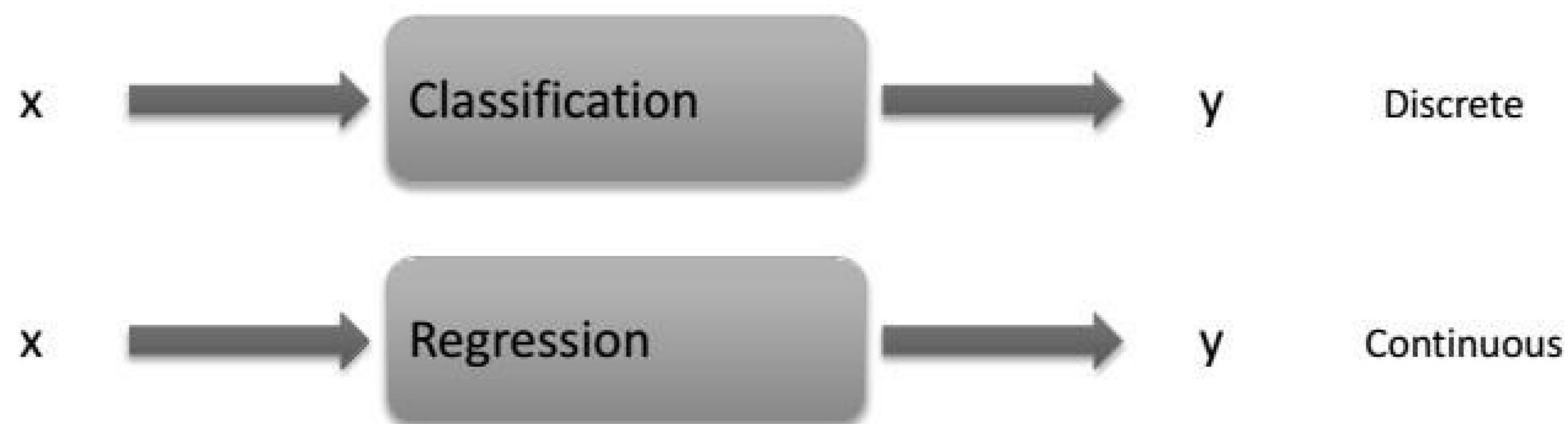
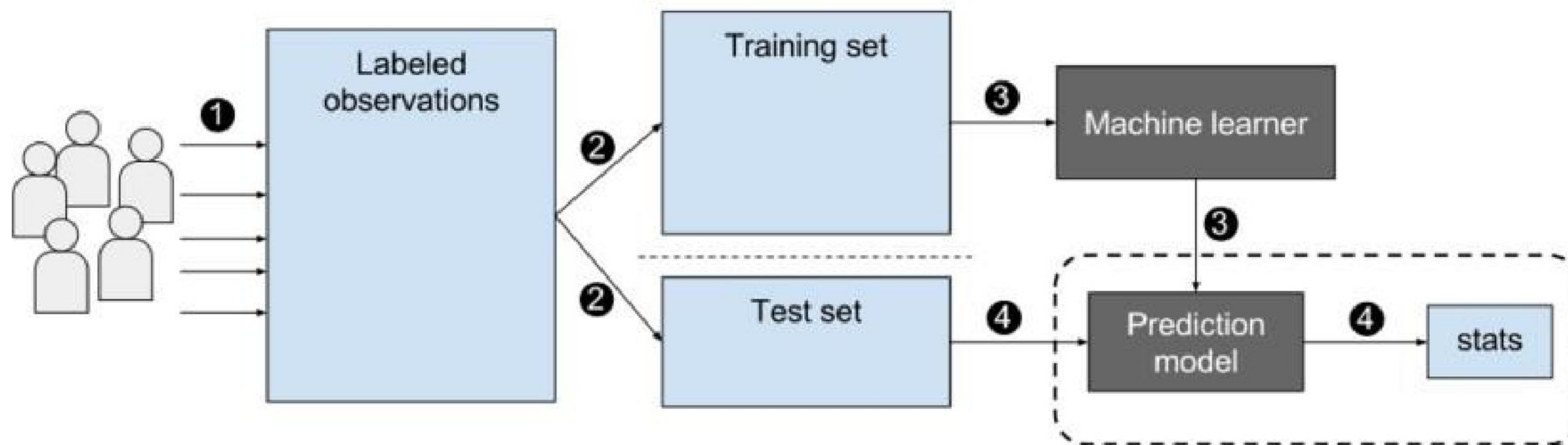
Assistant Professor



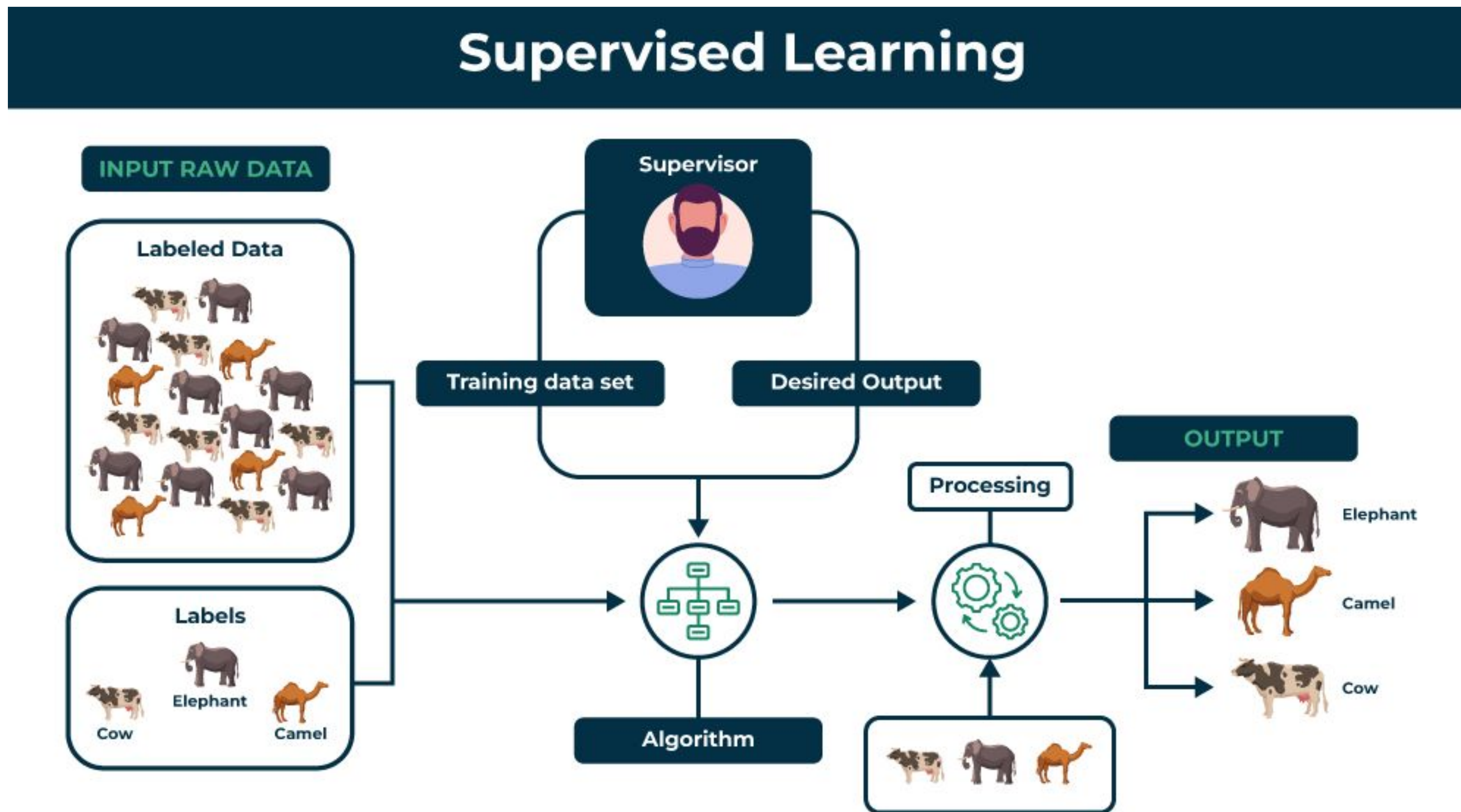
Types of Machine Learning



Supervised Learning



Supervised Learning



Supervised Learning

Advantages:

- High accuracy
- Interpretable
- Can be used in pre-trained models

Disadvantages:

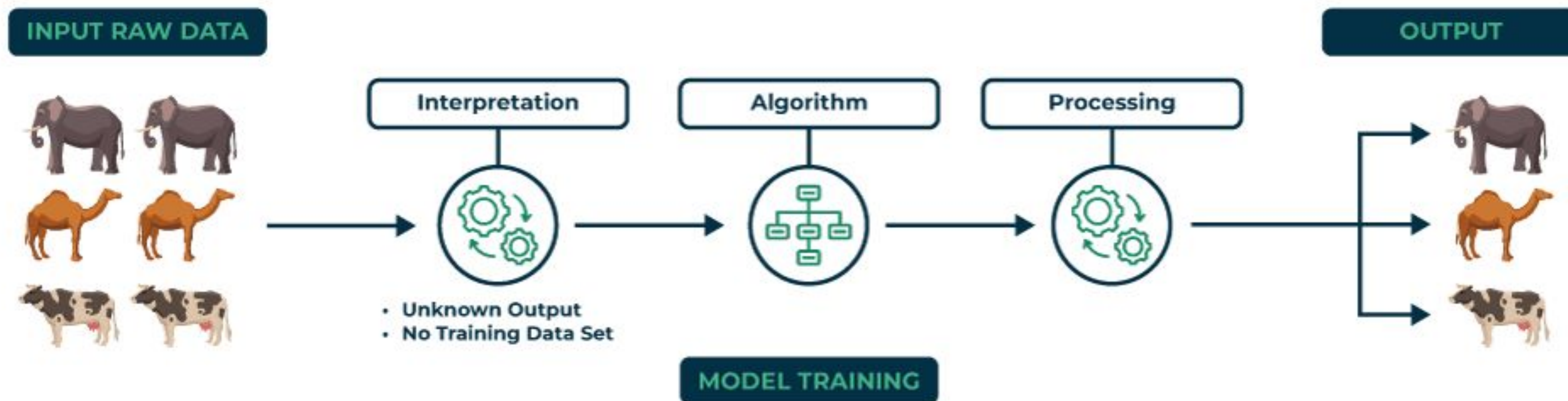
- May struggle with unseen or unexpected patterns
- Time-consuming and costly
- May lead to poor generalizations based on new data.

Applications:

Image classification, Natural language processing, Speech recognition, Medical diagnosis, Autonomous vehicles, Gaming, Weather forecasting

Unsupervised Learning

Unsupervised Learning



Two main types:

- Clustering
- Association

Unsupervised Learning

Advantages:

- Discovers hidden patterns and various relationships between the data.
- Used for tasks such as segmentation, anomaly detection, and data exploration.
- Reduces the effort of data labeling.

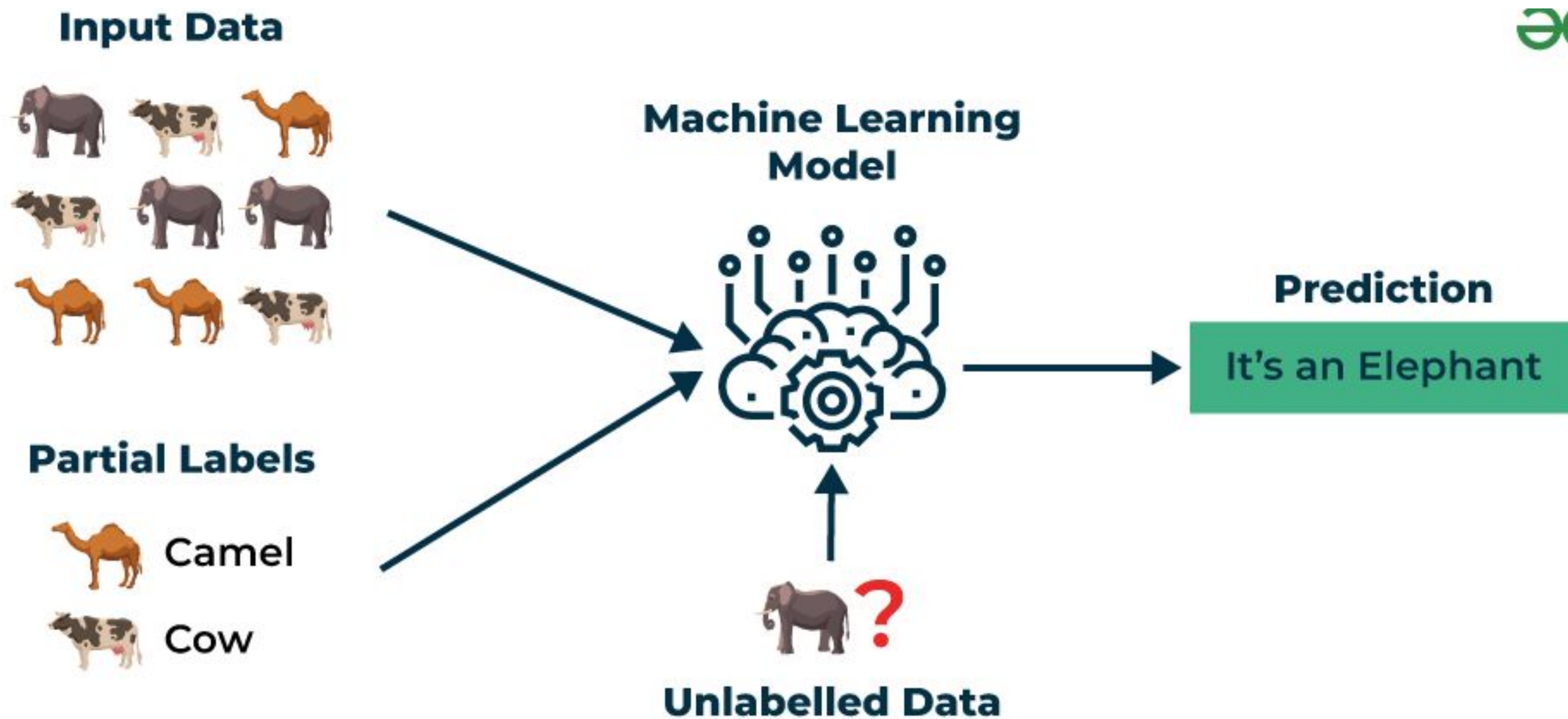
Disadvantages:

- Without using labels, it may be difficult to predict the quality of the model's output.
- Cluster Interpretability may not be clear and may not have meaningful interpretations.

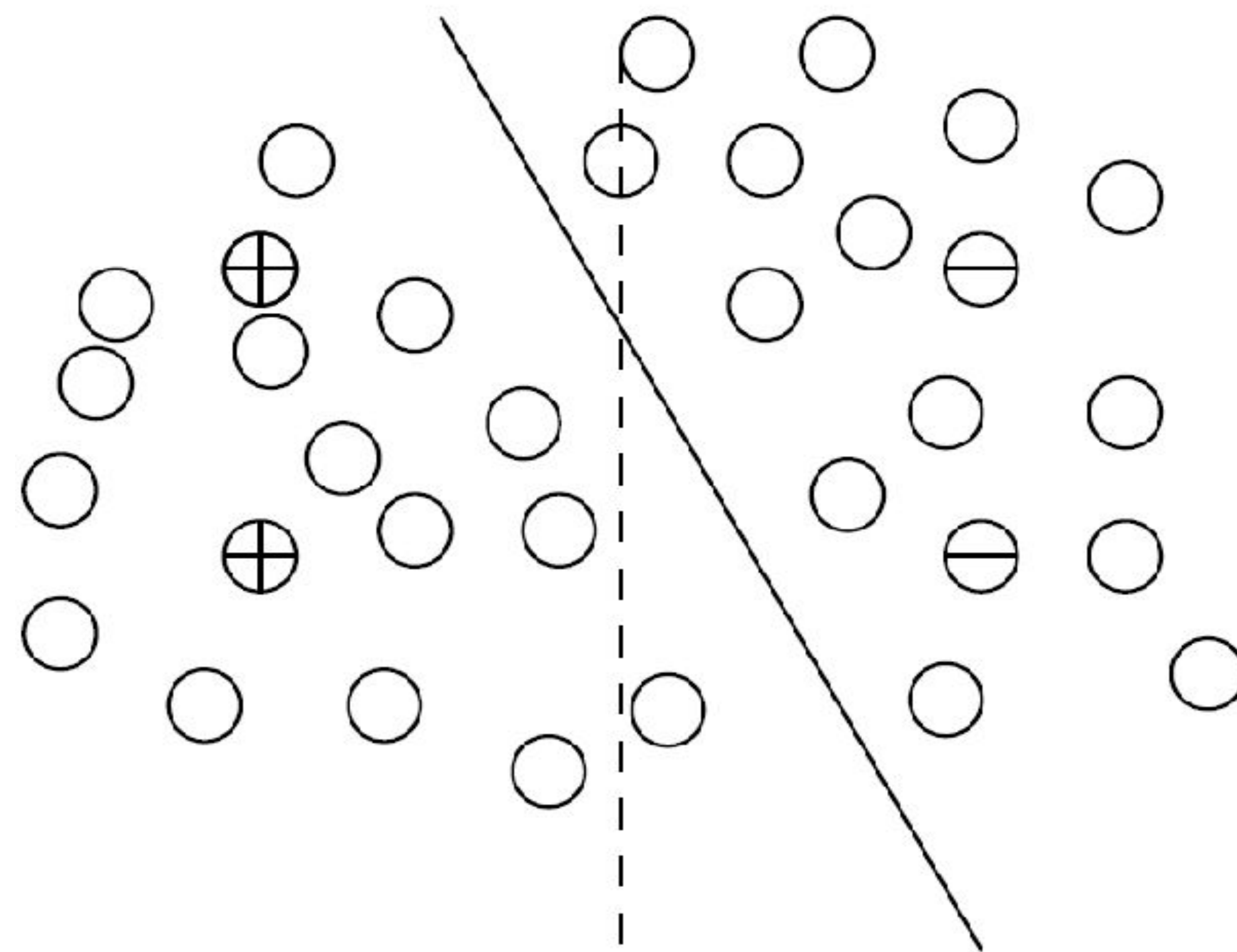
Applications:

Clustering, Anomaly detection, Dimensionality reduction, Image and video compression, Data preprocessing, Image segmentation, Customer behavior analysis.

Semi-Supervised Learning



Semi-Supervised Learning



- $\oplus$ Positive example - - - - Decision boundary without unlabeled examples
- $\ominus$ Negative example ———— Decision boundary with unlabeled examples
- $\bigcirc$ Unlabeled example

Semi-Supervised Learning

Advantages:

- It leads to better generalization as compared to supervised learning.
- Can be applied to a wide range of data.

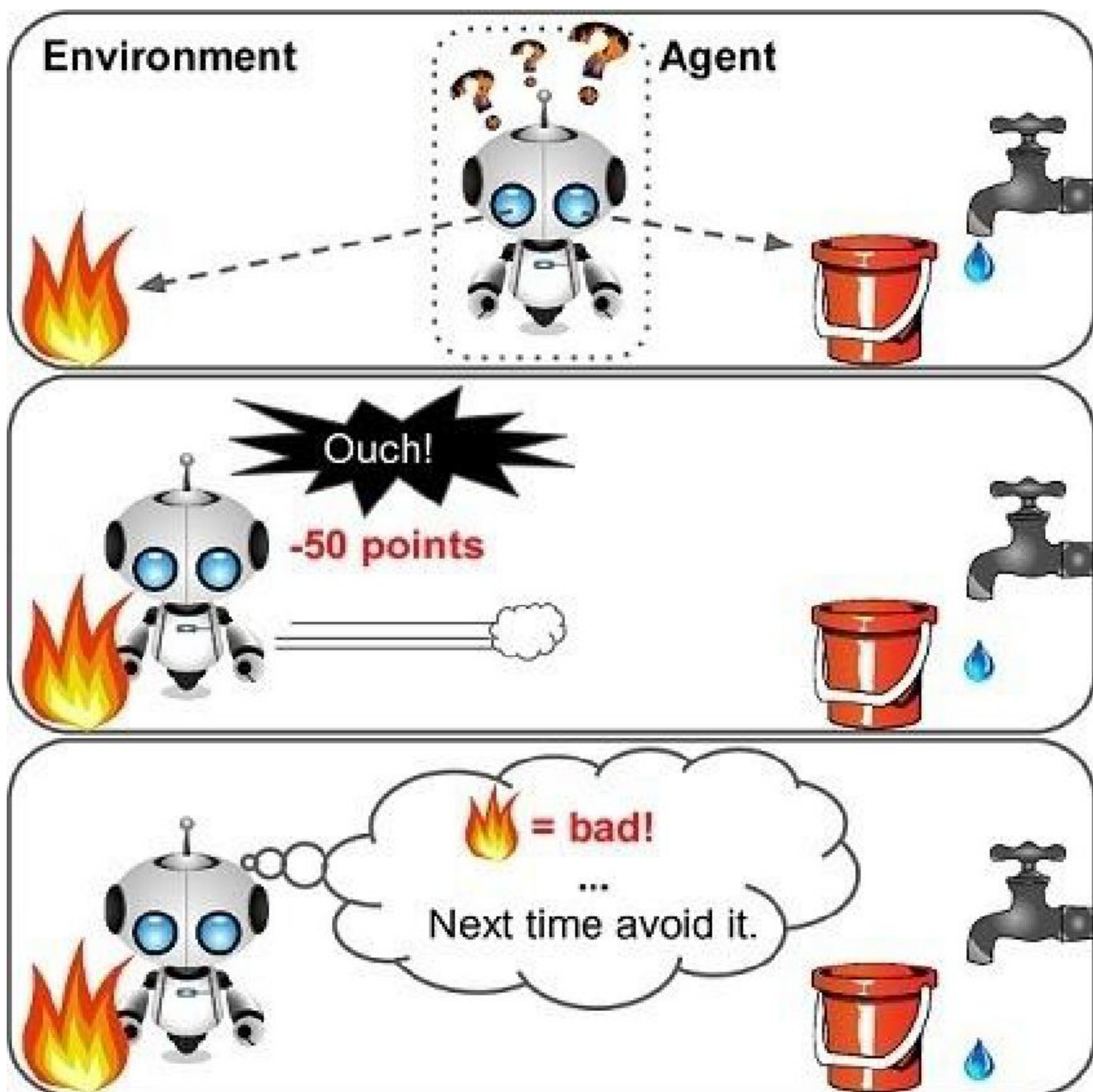
Disadvantages:

- Can be more complex to implement compared to other approaches.
- It still requires some labeled data that might not always be available or easy to obtain.
- The unlabeled data can impact the model performance accordingly.

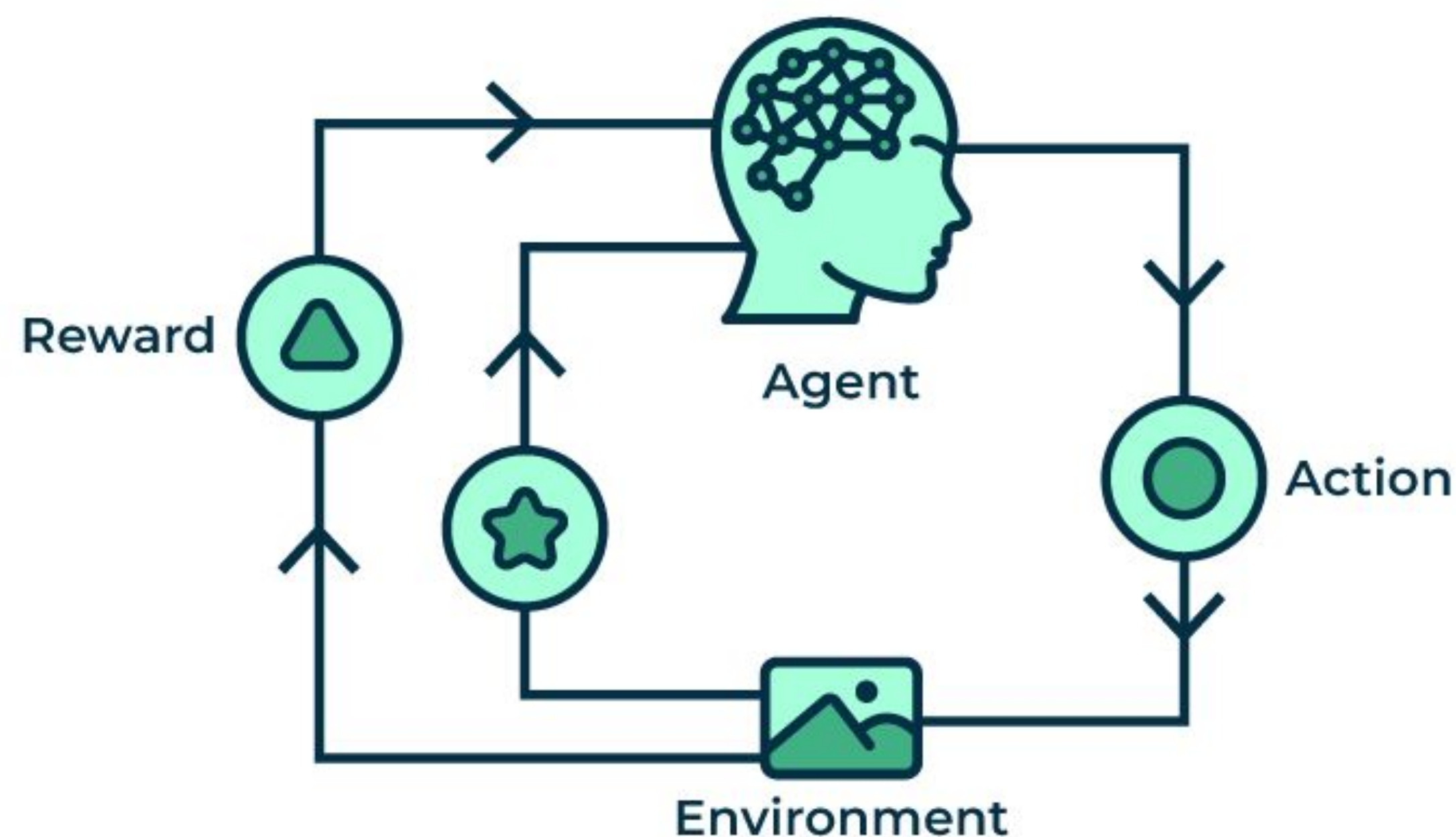
Applications:

Image Classification and Object Recognition, Healthcare and Medical Imaging

Reinforcement Learning



- 1 Observe
- 2 Select action using policy
- 3 Action!
- 4 Get reward or penalty
- 5 Update policy (learning step)
- 6 Iterate until an optimal policy is found



Reinforcement Learning

Advantages:

- It has autonomous decision-making that is well-suited for tasks where to make a sequence of decisions, like robotics and game-playing.
- Preferred to achieve long-term results that are very difficult to achieve.
- Used to solve complex problems that cannot be solved by conventional techniques.

Disadvantages:

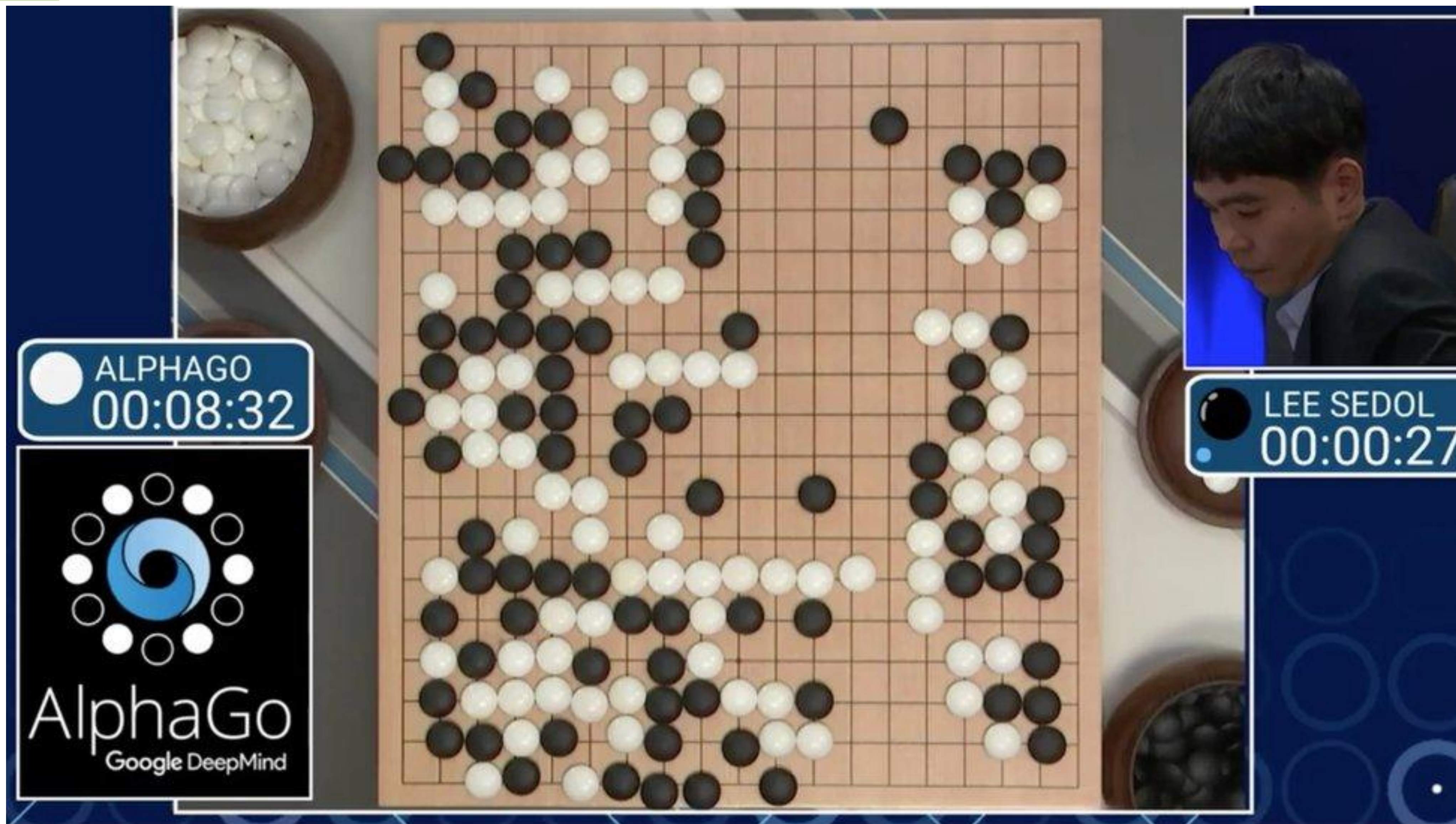
- Training agents can be computationally expensive and time-consuming.
- Not preferable to solving simple problems.
- Needs a lot of data and a lot of computation, which makes it costly.

Applications:

Game Playing, Robotics, Autonomous Vehicles, Healthcare, Adaptive Personal Assistants



Reinforcement Learning



Thank You

Any questions?

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